

COMPANION FOR CHAPTER 2

OF

**FUNDAMENTALS
OF PROBABILITY**

WITH STOCHASTIC PROCESSES

FOURTH EDITION

SAEED GHAHRAMANI

Western New England University
Springfield, Massachusetts, USA



CRC Press

Taylor & Francis Group

Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

A CHAPMAN & HALL BOOK

Contents

► **2 Combinatorial Methods** **3**

2A Additional Examples 3

Chapter 2

Combinatorial Methods

2A ADDITIONAL EXAMPLES

Example 2a One of Olivia's iPod playlists has only 12 songs, 6 by Lionel Richie and 6 by Adele. She sets the iPod to play the songs of that playlist. However, before doing that, Olivia unchecks and then rechecks the shuffle button of her iPod, and the iPod plays the playlist's songs in a random order, without repeating a song. What is the probability that no two songs of the same singer will be played back to back?

Solution: There are $12!$ ways for the songs to be played. In order that no two songs of the same singer be played back to back, the 1st, 3rd, 5th, 7th, 9th, and 11th songs the iPod will play should be Lionel Richie songs, and the 2nd, 4th, 6th, 8th, 10th, and 12th songs should be Adele songs, or vice versa. So the desired probability is

$$\frac{2 \times 6! \times 6!}{12!} = \frac{1}{462} \approx 0.00216. \quad \blacklozenge$$

Example 2b An 8-letter code is made by choosing distinct letters from the set of the English alphabet $\{a, b, c, \dots, z\}$. What is the probability that the letters a and b do not show up consecutively in either order?

Solution: The total number of possible 8-letter codes with distinct letters is ${}_{26}P_8$. Now we will count the number of codes in which the letters a and b show up consecutively in either order. When forming 8-letter codes, there are 7 ways in which b can immediately follow a , and 7 ways in which a can immediately follow b . For each of these 14 cases, there are ${}_{24}P_6$ possibilities for the remaining 6 letters of the 8-letter random code. So a and b can appear consecutively in either order in $14 \times {}_{24}P_6$ ways. Hence the desired probability is

$$1 - \frac{14 \times {}_{24}P_6}{{}_{26}P_8} = \frac{318}{325} \approx 0.98. \quad \blacklozenge$$

Example 2c An old lady claims that she can predict the sex of a baby by just looking at the shape of the face of the pregnant mother. At a Lamaze class, the lady meets 7 pregnant women and predicts the sexes of their children. Assuming that she does not have the ability that she claims she has, what is the probability that she predicts only 3 of the 7 sexes of the babies correctly?

Solution: Suppose that by $CCCIIII$ we mean that the lady predicts the sexes of the first 3 pregnant women she met correctly and the sexes of the remaining 4 pregnant ladies

incorrectly. We represent other cases similarly. The number of possibilities in which she is correct in exactly 3 cases is equal to the number of distinguishable sequences with three C 's and four I 's. So the probability that she predicts only 3 of the 7 sexes of the babies correctly is $\frac{7!}{3!4!}/2^7 \approx 0.273$. ♦

Example 2d In an international college, jewelry design is a very popular course, and Chaowei Yuan and 6 other Chinese students, 8 American, and 3 Indians are trying to take the course. However, there is room only for 5 more students to enroll. If the professor teaching the course chooses 5 of these students randomly and enrolls them, what is the probability that (a) Chaowei Yuan and exactly 2 of the 3 Indians are included? (b) Only one of the 8 American students is included?

$$\text{Solution: (a)} \quad \frac{\binom{3}{2}\binom{14}{2}}{\binom{18}{5}} = \frac{13}{408} \approx 0.0319.$$

$$\text{(a)} \quad \frac{\binom{8}{1}\binom{10}{4}}{\binom{18}{5}} = \frac{10}{51} \approx 0.196. \quad \blacklozenge$$

Example 2e In the Congress of the United States of America, regardless of population or size, each of the 50 states is represented by two senators. For an interview concerning an important pending legislation, a TV station chooses 8 senators at random for interview. What is the probability that no two of the selected senators are from the same state.

Solution: The TV station has $\binom{100}{8}$ choices for 8 senators to interview. If we want no two senators to be from the same state, first we need to choose 8 states from the 50, for which we have $\binom{50}{8}$ choices. Then for each of these $\binom{50}{8}$ choices, we have 2^8 ways to choose one senator from each of the chosen 8 states. So the desired probability is

$$\frac{\binom{50}{8} 2^8}{\binom{100}{8}} \approx 0.739. \quad \blacklozenge$$

Example 2f In a medical study, 25 patients with a duodenal ulcer have volunteered to be treated by an experimental drug. The researcher has decided to treat 18 of the patients, randomly selected, by the new drug and to give the remaining 7 a placebo as control. If the duodenal ulcers of 17 of these patients are caused by the bacterium *H. pylori*, and the rest by nonsteroidal anti-inflammatory drugs such as aspirin, what is the probability that the duodenal ulcers of at least 13 of the experiment group (the patients treated by the drug) are caused by the bacterium *H. pylori*?

Solution: The answer is
$$\sum_{i=13}^{17} \frac{\binom{17}{i} \binom{8}{18-i}}{\binom{25}{18}} = \frac{859}{2185} \approx 0.393. \quad \blacklozenge$$

Example 2g In the spring semester of 2016, Zelma asked her probability professor why she had never been included in a presidential poll conducted by media. The estimated population of the United States, at that time, was 323,000,000, of which approximately 219,000,000 were eligible to vote (those who were at least 18 years old). The professor calculated the probability that Zelma be included in a random sample of 1000, which was the average size of a poll:

$$\frac{\binom{1}{1} \binom{219,000,000-1}{999}}{\binom{219,000,000}{1000}} = \frac{1}{219,000} \approx 0.0000046.$$

Based on this, he concluded that the chances were extremely small for Zelma to be included in a presidential random sample of size 1000. $\blacklozenge$

Example 2h By mistake, a student who is running a computer program enters with negative signs two of the six positive numbers and with positive signs two of the four negative numbers. If at some stage the program chooses three distinct numbers from these 10 at random and multiplies them, what is the probability that at that stage no mistake will occur?

Solution:
$$\frac{\binom{6}{3} + \binom{6}{1} + \binom{6}{1} + \binom{6}{1} \binom{2}{1} \binom{2}{1}}{\binom{10}{3}} = 0.467.$$

Another Solution:
$$\frac{\binom{6}{3} + \binom{6}{1} \binom{4}{2}}{\binom{10}{3}} = 0.467. \quad \blacklozenge$$

Example 2i According to the 1998 edition of *Encyclopedia Britannica*, “there are at least 15,000 to as many as 35,000 species of orchids.” These species have been found naturally and are distinct from each other. Suppose that hybrids can be created by crossing any two existing species. Furthermore, suppose that hybrids themselves can be continued to be hybridized with each other or with an original species. Orchid lovers develop thousands and thousands of hybrids for use as garden or greenhouse ornamental and for the commercial flower trade. Suppose that all species are crossed, two at a time, to create the first generation of hybrids. Then the first generation of hybrids are crossed with each other and with the original species, two at a time, to develop the second generation of hybrids. The second generation of hybrids are crossed with each other, with the first generation hybrids, and with the original species, two at a time, to generate the third generation of hybrids, and so on. Let n be the total number of original species of orchids. Let n_k be the number of hybrids in the k th generation.

- (a) Find n_k in terms of $n, n_1, \dots, n_{k-1}$.
- (b) For $n = 25,000$, find the largest possible total number of all hybrids in the first four generations.

Solution: (a) It must be clear that

$$\begin{aligned}
 n_1 &= \binom{n}{2} \\
 n_2 &= \binom{n_1}{2} + nn_1 \\
 n_3 &= \binom{n_2}{2} + n_2(n + n_1) \\
 n_4 &= \binom{n_3}{2} + n_3(n + n_1 + n_2) \\
 &\vdots \\
 n_k &= \binom{n_{k-1}}{2} + n_{k-1}(n + n_1 + \dots + n_{k-1}).
 \end{aligned}$$

- (b) For $n = 25,000$, successive calculations of n_k 's yield,

$$\begin{aligned}
 n_1 &= 312,487,500, \\
 n_2 &= 48,832,030,859,381,250, \\
 n_3 &= 1,192,283,634,186,401,370,231,933,886,715,625, \\
 n_4 &= 710,770,132,174,366,339,321,713,883,042,336,781,236, \\
 &\quad 550,151,462,446,793,456,831,056,250.
 \end{aligned}$$

For $n = 25,000$, the total number of all possible hybrids in the first four generations, $n_1 + n_2 + n_3 + n_4$, is 710,770,132,174,366,339,321,713,883,042,337,973,520,184,337,863,865,857,421,889,665,625. This number is approximately 710×10^{63} . ♦

Example 2j In Maryland's lottery, players pick six different integers between 1 and 49, the order of selection being irrelevant. The lottery commission then randomly selects six of these as the *winning numbers*. What is the probability that at least two consecutive integers are selected among the winning numbers?

Hint: Let $\mathcal{A}$ be the set of all 6-element combinations of $\{1, 2, \dots, 49\}$ with no consecutive integers. Let $\mathcal{B}$ be the set of all 6-element combinations of $\{1, 2, \dots, 44\}$. Begin by showing that there is a one-to-one correspondence between $\mathcal{A}$ and $\mathcal{B}$.

Solution: Let a 6-element combination of a set of integers be denoted by $\{a_1, a_2, \dots, a_6\}$, where $a_1 < a_2 < \dots < a_6$. It can be easily verified that the function $h: \mathcal{B} \rightarrow \mathcal{A}$ defined by

$$h(\{a_1, a_2, \dots, a_6\}) = \{a_1, a_2 + 1, \dots, a_6 + 5\}$$

is one-to-one and onto. Therefore, there is a one-to-one correspondence between $\mathcal{B}$ and $\mathcal{A}$. This shows that the number of elements in $\mathcal{A}$ is $\binom{44}{6}$. Thus the probability that no consecutive integers are selected among the winning numbers is $\binom{44}{6} / \binom{49}{6} \approx 0.505$. This implies that the probability of at least two consecutive integers among the winning numbers is approximately $1 - 0.505 = 0.495$. Given that there are 47 integers between 1 and 49, this high probability might be counter-intuitive. Even without knowledge of expected value, a keen student might observe that, on the average, there should be $(49 - 1)/7 = 6.86$ numbers between each a_i and a_{i+1} , $1 \leq i \leq 5$. Thus he or she might erroneously think that it is unlikely to obtain consecutive integers frequently. ♦

Example 2k In how many ways can 10 different photographs be placed in six different envelopes, no envelope remaining empty?

Hint: An easy way to do this problem is to use the following version of the inclusion-exclusion principle: Let $A_1, A_2, \dots, A_n$ be n subsets of a finite set Ω with N elements. Let $N(A_i)$ be the number of elements of A_i , $1 \leq i \leq n$, and $N(A_i^c) = N - N(A_i)$. Let S_k be the sum of the elements of all those intersections of $A_1, A_2, \dots, A_n$ that are formed of exactly k sets. That is,

$$\begin{aligned} S_1 &= N(A_1) + N(A_2) + \dots + N(A_n), \\ S_2 &= N(A_1 A_2) + N(A_1 A_3) + \dots + N(A_{n-1} A_n), \end{aligned}$$

and so on. Then

$$N(A_1^c A_2^c \dots A_n^c) = N - S_1 + S_2 - S_3 + \dots + (-1)^n S_n.$$

Solution: Let N be the number of ways that 10 different photographs can be placed in six different envelopes, allowing for the possibility of empty envelopes. Let A_i be the set of all situations in which envelope i is empty. Then the desired quantity is $N(A_1^c A_2^c \dots A_6^c)$. Clearly, $N = 6^{10}$, $N(A_i) = 5^{10}$, $N(A_i A_j) = 4^{10}$, $i \neq j$, and so on. So S_1 has $\binom{6}{1}$ equal terms, S_2 has $\binom{6}{2}$ equal terms, and so on. Therefore, the solution is

$$6^{10} - \binom{6}{1} 5^{10} + \binom{6}{2} 4^{10} - \binom{6}{3} 3^{10} + \binom{6}{4} 2^{10} - \binom{6}{5} 1^{10} + \binom{6}{6} 0^{10} = 16,435,440. \quad \blacklozenge$$

Example 2ℓ We are given n ($n > 5$) points in space, no three of which lie on the same straight line. Let Ω be the family of planes defined by any three of these points. Suppose that the points are situated in a way that no four of them are coplanar, and no two planes of Ω are parallel. From the set of the lines of the intersections of the planes of Ω , a line is selected at random. What is the probability that it passes through none of the n points?

Hint: For $i = 0, 1, 2$, let A_i be the set of all lines of the intersections that are determined

by planes having i of the given n points in common. If $|A_i|$ denotes the number of elements of A_i , the answer is $|A_0|/(|A_0| + |A_1| + |A_2|)$.

Solution: For $i = 0, 1, 2$, let A_i be the set of all lines of the intersections that are determined by planes having i of the given n points in common. If $|A_i|$ denotes the number of elements of A_i , the answer is $|A_0|/(|A_0| + |A_1| + |A_2|)$. $|A_0| = \frac{1}{2} \binom{n}{3} \binom{n-3}{3}$,

$$|A_1| = \frac{1}{2} \binom{n}{3} \binom{3}{1} \binom{n-3}{2}, \quad |A_2| = \frac{1}{2} \binom{n}{3} \binom{3}{2} \binom{n-3}{1}.$$

The answer is

$$\frac{|A_0|}{|A_0| + |A_1| + |A_2|} = \frac{(n-4)(n-5)}{n^2 + 2}. \quad \blacklozenge$$